

# Comparative Assessment: INJONGO vs. MASSIVE

Deployment context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

## Deployment Context

**Use case:** Amharic intercity-travel booking bots for Sky Bus, Selam Bus, and Ride covering routes, fares, and seat changes

**Target population:** Inter-regional commuters and travelers booking long-distance buses and ride-hail trips via smartphone apps

## Overall Risk Ratings

| Benchmark | Risk | Avg. Score |
|-----------|------|------------|
| INJONGO   | HIGH | 3.0 / 5    |
| MASSIVE   | HIGH | 2.3 / 5    |

## Dimension Scores Comparison

| Dimension            | INJONGO | Rating           | MASSIVE | Rating           | Δ    |
|----------------------|---------|------------------|---------|------------------|------|
| Input Ontology (IO)  | 2/5     | Significant gaps | 2/5     | Significant gaps | 0    |
| Input Content (IC)   | 3/5     | Moderate gaps    | 1/5     | Serious concern  | +2   |
| Input Form (IF)      | 5/5     | Strong alignment | 4/5     | Minor gaps       | +1   |
| Output Ontology (OO) | 2/5     | Significant gaps | 2/5     | Significant gaps | 0    |
| Output Content (OC)  | 2/5     | Significant gaps | 1/5     | Serious concern  | +1   |
| Output Form (OF)     | 4/5     | Minor gaps       | 4/5     | Minor gaps       | 0    |
| Average              | 3.0     |                  | 2.3     |                  | +0.7 |

Δ = INJONGO score – MASSIVE score. Positive = regional benchmark scores higher.

## Overall Summaries

### INJONGO (risk: HIGH)

INJONGO is the most relevant existing benchmark for Amharic intent detection and slot-filling, and the only large-scale resource of its kind. For the Ethiopian intercity/ride-hail deployment, it provides genuine value in three respects: (1) Amharic is included as a first-class language with culturally grounded entities (Ethiopian banks, TeleBirr, Ethiopian cities including Dire Dawa, Bahir Dar, Hawassa, Arba Minch, and even Ge'ez calendar months Meskerem and Hamle as DATE slot values); (2) text/Ge'ez script and structured intent+slot output exactly match deployment requirements; (3) annotation quality after review is high ( $\kappa=0.933$  for Amharic). However, three substantive gaps limit construct validity for this specific deployment: (a) the 40-intent travel taxonomy is air-travel-framed and entirely lacks intercity bus, seat-class, fare-tier, and operator-specific rebooking intents — confirmed empirically in the Amharic split (HIGH-priority IO concern); (b) Ethiopian Birr is absent as a CURRENCY/MONEY slot value in the sampled Amharic data while other tracks' currencies (Naira, Rand) are well-represented, and most Amharic DATE slots use relative expressions rather than named Ge'ez months (HIGH-priority IC residuals); (c) no code-switched Amharic–Oromo / Amharic–Tigrinya utterances exist despite the L2-heavy multi-regional user base, and annotator regional demographics are undisclosed (HIGH-priority OC concern). The benchmark is suitable as a foundation but requires deployment-specific supplementation before being treated as sufficient evaluation evidence.

## MASSIVE (risk: HIGH)

MASSIVE provides a strong modality and output-format match for an Amharic intent-classification + slot-filling booking bot, but exhibits severe content-level gaps for the Ethiopian intercity-travel deployment. The translation-from-English pipeline [Q2, Q4, Q106] guarantees that Amharic data lacks Ethiopian calendar dates, local city name variants, Birr fare conventions, and code-switching with Oromo or Tigrinya — all of which the user has confirmed appear in real user utterances. Operator-specific intents and slot types (operator\_name, seat\_class, route\_closure\_reason, surcharge\_event, departure\_terminal) are absent from the 60-intent / 55-slot taxonomy, which the paper itself describes as 'relatively simple' [Q108]. Annotator demographics for Amharic are undisclosed, raising convergent-validity concerns about whether labels reflect Ethiopian intercity-traveler judgments. Per-language data is small (~19.5k records, [Q103]) and Amharic — a unique-script, lower-resource language — is expected to fall in the lower performance tier per the script/pretraining-data correlation [Q87, Q98, Q100]. Strengths: text-only modality match, Ethiopic-script inclusion, output format directly compatible, and rich per-language breakdowns enabling expected-performance estimation.

## Per-Dimension Justification Comparison

Each benchmark's full assessment justification and cited evidence, shown side by side.

### Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

| INJONGO — 2/5 (Significant gaps) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MASSIVE — 2/5 (Significant gaps) [medium confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| INJONGO's travel domain is derived from CLINC's generic intent set ('exchange rate', 'book flight') [Q39] and explicitly omits intercity bus operator flows. Empirical inspection of the Amharic split confirms that all sampled travel-domain Amharic utterances use air-travel framing (book_flight to Turkey, Sudan) or generic car rental — there are no intercity bus, seat-class, or operator-specific intents [DATASET-D7, D8, D25, D26]. The deployment requires intent detection for Sky Bus / Selam Bus / Ride bus and ride-hail booking, including seat-class upgrade, road-closure rebooking, holiday-surge booking, and operator-specific cancellation, none of which appear in the 40-intent taxonomy. The paper itself acknowledges scope limits [Q107, Q111]. Although elicitation marked IO as MODERATE priority (user accepts general travel intents as broadly comparable), the empirical ontology mismatch in the travel subdomain is concrete and substantial. | MASSIVE's taxonomy of 18 domains, 60 intents, and 55 slots [Q1] was designed around major virtual assistants [Q16] and the 'general multi-domain NLU task' [Q12]. The corpus consists primarily of interrogatives and imperatives [Q30], which broadly aligns with booking-bot utterances. However, the taxonomy contains no intercity-transport-specific intents (route_query, operator_query, seat_class_query, cancel_and_rebook for road closures, holiday_surcharge_query, departure_terminal_query) that are explicitly required for the deployment, and the paper acknowledges its labeling schema is 'relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options' [Q108]. The user views booking flows as broadly generic (Q1 elicitation), reducing concern, but at least six intent types relevant to the deployment are absent, and a confirmed full gap exists across African NLP for transport-domain NLU intents (search_target gap_id 6). |
| INJONGO evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | MASSIVE evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [Q39] 'we extracted 40 intents from five most suitable domains to the African contexts: Banking ... Travel (e.g. "exchange rate", "book flight")' (p.3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | [Q1] 'MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots.' (p.1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| [Q107] 'The scope of INJONGO is constrained by its coverage of only 5 domains and 40 intents, missing some other domains such as healthcare and education that are essential for real-world applications.' (p.9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | [Q12] 'This work focuses on the general multi-domain NLU task and builds off the SLURP (Bastianelli et al., 2020) benchmark dataset to extend to an unprecedented 50 new languages.' (p.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| [Q111] 'the fixed distribution of examples across intents may not accurately reflect the natural frequency of these interactions in real-world conversations.' (p.9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | [Q16] 'Second, we determined existing languages available in major virtual assistants, such that the dataset could be used to benchmark today's systems.' (p.3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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| <a href="#">DATASET-D7</a> : Amharic book_flight is international air travel (to Turkey), no surface transport       | <a href="#">[Q30]</a> 'Specifically, the corpus primarily consists of interrogatives (i.e., questions) and imperatives (commands or requests).' (p.4)                                 |
| <a href="#">DATASET-D8</a> : Amharic book_flight to Sudan — international air, no bus operator                       | <a href="#">[Q108]</a> 'Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options.' (p.9) |
| <a href="#">DATASET-D25, D26</a> : Both Amharic book_flight examples are international air; no intercity bus intents |                                                                                                                                                                                       |

## Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

| INJONGO — 3/5 (Moderate gaps) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | MASSIVE — 1/5 (Serious concern) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <p>Empirical dataset inspection partially resolves what the paper does not document. Ethiopian calendar months (Meskerem, Hamle) appear as DATE slot values <a href="#">[DATASET-D1, D2, D6]</a>, Ethiopian intercity cities (Dire Dawa, Bahir Dar, Hawassa, Arba Minch) appear as CITY_OR_PROVINCE slots <a href="#">[DATASET-D2, D9]</a>, and Ethiopia-specific financial entities (Amhara Bank, Commercial Bank of Ethiopia, TeleBirr, Dashen Bank) appear in banking-domain instances <a href="#">[DATASET-D3, D4, D5, D34]</a>. This is materially better than the registry feared. However, several deployment-critical content gaps remain: (a) Ethiopian Birr (ETB) was absent as a CURRENCY/MONEY slot value in any sampled Amharic example despite Naira/Rand appearing in other tracks <a href="#">[DATASET-D5, D19, D22 vs Amharic absence]</a>; (b) most Amharic DATE slots used relative expressions rather than named Ge'ez months (only 2/9) <a href="#">[DATASET-D15, D16, D28]</a>; (c) Ethiopian time convention (6-hour offset) is unaddressed in TIME annotations <a href="#">[DATASET-D11, D23]</a>; (d) no code-switched Amharic–Oromo / Amharic–Tigrinya utterances appear, despite ~70% L2 Amharic users in the deployment population <a href="#">[DATASET-D1–D18 monolingual]</a>. IC was marked HIGH priority in elicitation, so these residual gaps weigh heavily.</p> | <p>MASSIVE's Amharic content was created by 'tasking professional translators to localize the English-only SLURP dataset' <a href="#">[Q2, Q4]</a>, with seed data 'originally created through crowd-sourcing, not from a real virtual assistant, which introduces artificialities' <a href="#">[Q106]</a>. This means Amharic instances will not contain Ethiopian calendar dates (Meskerem, Tikimt — 13-month Ge'ez calendar offset 7–8 years from Gregorian, <a href="#">[WEB-4]</a>), local city name variants (Adama/Nazret, Mekelle/Mek'ele), Birr-denominated fares, or holiday-surcharge contexts (Timkat, Meskel, Enkutatash — <a href="#">[WEB-5, WEB-6]</a>) — all of which the user has confirmed are required slot values (Q2 elicitation). Worker-decided localization 'added further noise' <a href="#">[Q107]</a>, and per-language data is 'relatively small at 19.5k total records and 11.5k records for training' <a href="#">[Q103]</a>. The IC dimension was rated HIGH priority and the mismatch is severe.</p> |
| INJONGO evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MASSIVE evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <a href="#">DATASET-D1, D2</a> : 'ከድሬዳዋ ወደ ባህርዳር ለመስከረም እምስት' — Ethiopian cities + Ge'ez month Meskerem in travel-domain DATE slot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <a href="#">[Q2]</a> 'MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera.' (p.1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <a href="#">DATASET-D6</a> : Hamle (Ethiopian month) used in DATE slot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <a href="#">[Q4]</a> 'MASSIVE was created by localizing the SLURP NLU dataset (created only in English) in a parallel manner.' (p.1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <a href="#">DATASET-D9</a> : Hawassa and Arba Minch as CITY_OR_PROVINCE slots in travel_suggestion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <a href="#">[Q103]</a> 'Starting with the dataset, the per-language data quantities are relatively small at 19.5k total records and 11.5k records for training.' (p.9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <a href="#">DATASET-D3, D4, D5</a> : Amhara Bank, Commercial Bank of Ethiopia, TeleBirr — Ethiopia-specific banking entities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <a href="#">[Q106]</a> 'Third, the data were originally created through crowd-sourcing, not from a real virtual assistant, which introduces artificialities.' (p.9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| <a href="#">DATASET-D15, D16, D28</a> : Most Amharic DATE annotations use relative expressions ('this month', 'next month') rather than named Ethiopian calendar months                                               | <a href="#">[Q107]</a> 'Relatedly, allowing the worker to decide on translation versus localization of slot entities added further noise to the dataset, although we try to store this decision in the metadata.' (p.9) |
| <a href="#">DATASET-D5</a> vs <a href="#">D19, D22</a> : TeleBirr present as PAYMENT_COMPANY but no Birr-denominated CURRENCY/MONEY slot in Amharic sample, while Naira (Hausa) and Rand (Xhosa) are well-represented | <a href="#">[WEB-4]</a> Ethiopian (Ge'ez) calendar: 13 months, 7–8-year offset from Gregorian — temporal expression slot values absent from MASSIVE                                                                     |
| <a href="#">DATASET-D1–D18</a> : All Amharic utterances monolingual; no Oromo/Tigrinya/English code-switching observed                                                                                                | <a href="#">[WEB-5]</a> Timkat/Ethiopian Epiphany — major travel surge, Tir 11 E.C.                                                                                                                                     |
| <a href="#">DATASET-D11, D23</a> : Ethiopian clock convention (12 ሰዓት) ambiguity; one example unannotated                                                                                                             | <a href="#">[WEB-6]</a> Meskel, Genna, Enkutatash — peak travel periods unrepresented in MASSIVE                                                                                                                        |

## Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

| INJONGO — 5/5 (Strong alignment) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MASSIVE — 4/5 (Minor gaps) [medium confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| INJONGO is a text-only benchmark <a href="#">[Q38, Q43]</a> released under CC BY 4.0 <a href="#">[Q23]</a> , with Amharic encoded in Ge'ez/Ethiopic script. The deployment is also a text-based smartphone app, with primary input in Ge'ez script. Dataset inspection confirms all sampled Amharic examples use proper Ge'ez script with no romanization or mixed-script anomalies <a href="#">[DATASET-D1, D5]</a> . There is full modality and script alignment. Elicitation marked IF as LOWER priority, consistent with no concrete violations. | Modality alignment is strong: MASSIVE is text-only <a href="#">[Q3]</a> and Amharic is included as one of 51 languages, with 21 distinct scripts represented <a href="#">[Q25]</a> including Ethiopic/Ge'ez fidel as a 'unique' script <a href="#">[Q27]</a> . The deployment is also text-only smartphone input in Ethiopic script. Script diversity was explicitly sought 'to drive experimentation in tokenization and normalization' <a href="#">[Q20]</a> . The paper acknowledges 'our collection system did not have a robust method to preserving or denoting native tokenization practices' <a href="#">[Q109]</a> , leaving slot-boundary handling for Amharic's morphologically rich, fusional structure as an open question (gap_id 8 PARTIAL — amseg and HornMorpho exist but not evaluated against MASSIVE Amharic). Amharic uses spaces as word delimiters, so unlike Japanese/Chinese/Khmer it does not require character-level splitting <a href="#">[Q80, Q92, Q96]</a> . The IF dimension is rated LOWER priority and modality match is good, but documentation of Amharic-specific tokenization quality is absent. |
| INJONGO evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MASSIVE evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">[Q23]</a> 'Dataset is released under CC BY 4.0 license.' (p.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <a href="#">[Q3]</a> '587k train utterances, 104k dev utterances, 152k test utterances, and 153k utterances currently held out' (p.1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <a href="#">[Q38]</a> 'the dataset construction follows two stages: (1) Utterance elicitation in an African language and (2) Slot-filling annotation of the generated utterance.' (p.3)                                                                                                                                                                                                                                                                                                                                                              | <a href="#">[Q20]</a> 'Fifth, we examined the script of each language, seeking to increase script diversity to drive experimentation in tokenization and normalization.' (p.3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <a href="#">[Q43]</a> 'The annotation followed the named entity recognition setup on LabelStudio platform.' (p.4)                                                                                                                                                                                                                                                                                                                                                                                                                                    | <a href="#">[Q25]</a> 'There are 21 distinct scripts used in the dataset.' (p.3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">DATASET-D1</a> : Amharic utterance in proper Ge'ez script throughout                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">[Q27]</a> 'The Arabic script is used for three languages, the Cyrillic script for two languages, and the remaining 18 languages have "unique" scripts...' (p.3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <a href="#">DATASET-D5</a> : Ge'ez script consistent throughout Amharic banking example                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <a href="#">[Q109]</a> 'Fifth, our collection system did not have a robust method to preserving or denoting native tokenization practices...' (p.9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

| INJONGO — 2/5 (Significant gaps) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MASSIVE — 2/5 (Significant gaps) [medium confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <p>The 23-slot final ontology [Q118, Q119, Q144] includes structural fits for deployment needs (CITY_OR_PROVINCE, MONEY, CURRENCY, DATE, TIME, BANK_NAME), and merging of low-frequency types is principled [Q51, Q52, Q55]. However, several decision-rule mismatches arise for the Ethiopian deployment: (a) DATE/TIME boundary rules are documented [Q153, Q154] using Gregorian conventions only — no accommodation for Ethiopian calendar (13 months, Pagume intercalary) or 6-hour offset clock; (b) the 40-intent inventory excludes intercity bus, seat-class, fare-tier, and rebooking categories required for Sky Bus/Selam Bus/Ride [Q39, DATASET-D7, D8]; (c) dataset inspection shows Amharic DATE annotations dominated by relative expressions, suggesting the operative decision rule for Ethiopian month names is undertested [DATASET-D15, D16, D28]. These are structural-validity gaps. A redeeming factor: Ethiopian month names did appear as DATE slots [DATASET-D2, D6], so the ontology can host them in principle.</p> | <p>The output label space (60 intents and 55 slots [Q1]) is the same structure used to generate the input ontology, so the same gaps apply on the output side. Operator-specific slot types required by the deployment — operator_name, seat_class, route_closure_reason, surcharge_event, trip_change_type, departure_terminal — are absent from the 55-slot taxonomy (regional context, operator_specific_slot_types). The paper itself describes the schema as 'relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options' [Q108]. Output decision rules (label/F1/exact match) are language-agnostic, so cultural decision-rule misalignment per se is limited — the violation is primarily content-validity (missing categories) rather than structural-validity for the labels that do exist. The text-to-text decoder format ('sequence of labels ... followed by the intent' [Q73]) is compatible with the deployment.</p> |
| INJONGO evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | MASSIVE evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| [Q119] 'After merging similar or low-frequency types during data preprocessing in Section 3.3, it was reduced to 23 distinct slot types' (p.13)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | [Q1] '... 18 domains, 60 intents, and 55 slots.' (p.1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| [Q144] 'ACCOUNT_TYPE, ARTIST_NAME, BANK_NAME, BILL_TYPE, CALENDAR_EVENT, CITY_OR_PROVINCE, COUNTRY, CURRENCY, DATE, ... TIME' (p.19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | [Q73] 'The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent.' (p.7)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| [Q153] 'Anything that is less than one day should be annotated as TIME and not DATE' (p.21)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | [Q108] 'Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options.' (p.9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| [Q154] 'Anything that is more than one day must be annotated as DATE and not time' (p.21)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| DATASET-D2, D6: Ge'ez month names accepted as DATE slot values empirically                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| DATASET-D15, D16, D28: Most DATE slot values are relative expressions rather than named Ethiopian months                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

| INJONGO — 2/5 (Significant gaps) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | MASSIVE — 1/5 (Serious concern) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <p>Annotation methodology is sound on its face: <math>\geq 3</math> native annotators per utterance with majority voting [Q42, Q45], Fleiss' Kappa improving to 0.912–1.00 after a two-round review [Q47, Q48], with Amharic specifically reaching <math>\kappa=0.933</math> post-review (initial 0.836). However, several documented and empirical gaps reduce convergent validity for this deployment: (a) annotator regional/L2 demographics within Ethiopia are undisclosed [Q44, gap-4 NOT FOUND]; (b) the actual user population includes L2 Amharic speakers from Oromia, Tigray, and SNNP regions who code-switch — a population mismatch flagged HIGH in elicitation; (c) dataset inspection reveals concrete annotation gaps (empty spans on utterances containing clear DATE/TIME content [DATASET-D11, D33], inconsistent treatment of '12 ሰዓት' across alarm utterances [DATASET-D11 vs D23], misannotated PERSONAL_NAME in Oromo [DATASET-D29]); (d) no code-switched utterances at all in 19 Amharic samples [DATASET-D1-D18]. OC was marked HIGH priority in elicitation, so these residuals weigh heavily.</p> | <p>Annotators were vendor-pool MTurk workers selected via a listening-based fluency test [Q51], a three-judgment quality test [Q52], and a translator quiz with 'customized local-language examples' [Q53]; workers self-certified as 'native speaker or ... fluent in the required language' [Q122]. Annotator demographics — whether Amharic translators are urban Addis residents, Amhara-region speakers, or L2 speakers — are not disclosed (regional context, amharic_dialect_variation.massive_annotator_dialect_representation: NOT FOUND). The paper acknowledges 'the lack of labeled data ... that is realistic for the task and that is natural for each given language' [Q6] as the central problem the dataset attempts to address — but the translation-from-English pipeline means Amharic instances reflect English-language pragmatics translated into Amharic [Q2, Q4]. The user has confirmed code-switching with Oromo (Q3 elicitation) will occur in real utterances, and gaps 3 and 4 confirm no Oromo–Amharic or Tigrinya–Amharic code-switching booking-domain corpus exists. 'Some low-quality utterances' are acknowledged [Q104] and removable only at the cost of further reducing the already small per-language sample [Q103, Q105]. OC was rated HIGH priority and the violation is severe.</p> |
| INJONGO evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MASSIVE evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>[Q42] 'Each utterance was annotated by three annotators so that we could check for agreement in the slot annotations.' (p.4)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>[Q6] 'one difficulty in creating massively multilingual NLU models is the lack of labeled data for training and evaluation, particularly data that is realistic for the task and that is natural for each given language.' (p.1)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>[Q44] 'all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya.' (p.4)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>[Q51] '... an Amazon MTurk-hosted fluency test where workers listen to questions and statements in the relevant language and were evaluated using a multiple-choice questionnaire.' (p.6)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>[Q45] 'a majority voting system with a minimum of three annotators per sentence to resolve disagreements.' (p.4)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>[Q104] 'Second, there are some low-quality utterances, both in the seed data and in the translations.' (p.9)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p>[Q48] 'agreement scores improved markedly across all languages, reaching 0.912-1.00.' (p.4)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>[Q105] '... if a user does filtering based on these judgments, then the data size decreases even further.' (p.9)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>DATASET-D11: Amharic alarm utterance with empty spans despite containing 'ነገ' (tomorrow) and '12 ሰዓት' (12 o'clock) — annotation gap</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>[Q122] 'By clicking "SUBMIT", I also certify that I am a native speaker or am fluent in the required language' (p.17)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p>DATASET-D33: Amharic calendar_update with empty spans despite birthday calendar event content</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>DATASET-D23: Same utterance type annotated inconsistently across two examples — '12 ሰዓት' once unannotated, once with TIME slot</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>DATASET-D29: Oromo PERSONAL_NAME 'caalaa' likely misannotated from 'galgala caalaan' phrase</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |



**DATASET-D1–D18:** No code-switched Amharic utterances in any sample — annotator pool/methodology did not produce them

## Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

| INJONGO — 4/5 (Minor gaps) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MASSIVE — 4/5 (Minor gaps) [high confidence]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <p>Output is intent class (closed 40-class label) plus structured slot-tag spans in \$\$-delimited text [Q145, Q146], scored with accuracy and F1 [Q84]. This matches the deployment's task-oriented dialogue NLU output requirements directly. Best fine-tuned models reach 93.7% intent accuracy / 85.6% F1 [Q19, Q78], setting an upper bound; Amharic-specific scores are reported in Table 5 (per-language) but not extracted in registry — gap noted. One minor concern: the eng config used for cross-lingual transfer contains lossy transliterations of Ethiopian entities (e.g., 'Dashen' → 'dasion') [DATASET-D34], which slightly degrades cross-lingual transfer evaluation quality. OF was marked LOWER priority and substantively the form fits.</p> | <p>MASSIVE's output format — intent classification + token-level slot-fill sequence [Q70, Q71, Q73] — directly matches what an intent-and-slot booking system requires. Three primary metrics are reported: 'exact match accuracy, intent classification accuracy, and slot-filling F1 score' [Q5], with per-language breakdowns [Q126, Q127, Q128] and 95% confidence intervals [Q89]. Per-language Amharic results are reported, allowing direct estimation of expected performance. However, two structural concerns remain: (1) Pearson correlation of 0.54 between XLM-R pretraining data quantity and zero-shot exact match [Q87], combined with the finding that 'Germanic genera and Latin scripts performed the best overall' [Q98] and that low-pretraining-data languages like Icelandic underperformed [Q100], strongly suggests Amharic — a unique-script, lower-resource language — will fall in the lower-performance tier; (2) deployment may eventually require text-to-speech for accessibility (literacy ~51.8% in 2017, urban–rural gap [WEB-3, WEB-12, WEB-11]), but MASSIVE evaluates only text outputs [Q5]. OF was rated LOWER priority and overall format alignment is strong.</p> |
| INJONGO evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | MASSIVE evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| [Q19] 'fine-tuning baselines could reach an accuracy of 93.7% and F1-score of 85.6 for intent detection and slot-filling tasks respectively.' (p.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | [Q5] 'We also present modeling results on XLM-R and mT5, including exact match accuracy, intent classification accuracy, and slot-filling F1 score.' (p.1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| [Q84] 'Evaluation is based on accuracy and F1-score for ID and SF tasks.' (p.7)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | [Q73] 'The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent.' (p.7)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| [Q145] 'Please ensure that the entities match the listed types and that unstated entities should not be included in the response if no entities are found, return \$\$ only.' (p.19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | [Q87] '... Pearson correlation of 0.54 for exact match' (p.7)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| [Q146] 'Output: PERSONAL_NAME John \$\$ CITY_OR_PROVINCE Paris \$\$ MONEY 100 \$\$ RESTAURANT_NAME Awater' (p.19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [Q98] 'Germanic genera and Latin scripts performed the best overall...' (p.8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| DATASET-D34: 'Dashen Bank' transliterated as 'dasion bank' in eng config — cross-lingual transfer source quality concern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | [Q100] 'Icelandic was the lowest-performing Germanic language, likely due to a lack of pretraining data...' (p.8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | [Q126] 'Exact match accuracy by language for our three models using the full dataset and the zero-shot setup.' (p.20)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | [WEB-3] Ethiopia adult literacy ~51.8% (2017)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | [WEB-12] UNESCO literacy data corroborates ~51.8% figure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | [WEB-11] Significant urban–rural literacy gap; pastoralist regions much lower                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |